

Qtangl Q-Day Readiness Report

Target: api.regionalbank.example

Scenario: bank-tls-inventory

Readiness score: 0.0/100

Coverage confidence: 56.0%

Assets discovered: 0

Remediation items: 0

Mosca inequality assessment (harvest-now-decrypt-later)

Formula: $X + Y > Z$ — sum $X+Y = 15.0$ yr, $Z = 12.0$ yr

HNDL risk level: moderate (inequality holds: False)

Mosca inequality holds: $X+Y$ (15.0 yr) $>$ Z (12.0 yr). Sensitive data may be decrypted after Q-Day if not migrated

Under these assumptions, migration can complete before Q-Day; HNDL timeline pressure is moderate but inverted

Compliance framework mapping

Regional bank TLS inventory — compliance pack

Mandate: NSM-10 / NIST IR 8547 PQC migration program

NSM-10 / NIST IR 8547: Federal PQC migration mandate for agencies and regulated financial partners.

PCI-DSS 4.0: Cardholder data environments require strong cryptography and key lifecycle controls.

NIST Cybersecurity Framework: Identify and protect cryptographic assets supporting customer-facing services.

PCI-DSS 4.0 (4.2 / 6.3): Protect cardholder data in transit with approved algorithms and key sizes.

NIST IR 8547 (Migration planning): Inventory quantum-vulnerable TLS endpoints and prioritize hybrid/PQC migration.

NSM-10 (Agency alignment): Track RSA/ECDSA exposure against 2030 deprecation timelines.

[gap] 1 asset(s): Module-Lattice-Based Digital Signature Algorithm standard

[gap] 1 asset(s): Transitioning to post-quantum cryptography standards

[gap] 1 asset(s): Commercial National Security Algorithm Suite 2.0

[gap] 1 asset(s): Cryptographic agility and key management requirements

[gap] 1 asset(s): Federal contractor cryptographic inventory expectations

Top remediation backlog

Post-quantum TLS handshake proof appendix

Mode: replayed | Server: test.openquantumsafe.org:4433

TLS: | Hybrid group:

KEM: | Groups:

ClientHello excerpt (2 hex chars): 00...

Captured TLS 1.3 ClientHello negotiating hybrid X25519MLKEM768 key exchange via Open Quantum Safe test server. (Liv